



Explainable deep learning architecture for early diagnosis of Parkinson's disease

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Abstract

Parkinson's disease (PD) is a neurodegenerative disease that develops in middle-aged and older adults. The development of a gait detection for PD patients to assist doctors in diagnoses is a crucial research target. This work develops an explainable learning architecture that involves deep learning, machine learning, data selection, feature evaluation and data balancing mechanisms, for gait detection in PD patients. The results obtained for these patients and healthy individuals are analyzed with different algorithms. Application of the research architecture that is developed in this study significantly increased the overall rate of identification; specifically, when used proposed architecture with XGBoost and deep learning achieves accuracy rates as high as 97.32% and 98.41%, respectively. The goal is to provide an early detection system that can serve as a reference for doctors when making a diagnosis.

Keywords Explainable deep learning · Gait detection · Parkinson's disease

1 Introduction

PD is a neurodegenerative disease that develops in middle-aged and older adults. Common symptoms of PD include tremor, slowed movement (Bradykinesia), rigid muscles, impaired posture and balance, gait disturbance, loss of automatic movement, changes in handwriting and changes in speech (Hoehn and Yahr 1998; Hirose et al. 2019; Ostrowski 2018). No cure currently exists for disease, which can only be alleviated through surgery and medication. However, all medications have side effects that can affect people's every-

day lives. There are symptoms including hallucinations, depression, sleep disorders and low blood pressure.

In clinical diagnosis, the symptoms of PD are associated with five stages. Stages 1 and 2 are milder than other stages, allowing patients to maintain their daily lives. But patients of stages 4 and 5 that are not able to get up or move on their own will be cared by other people (Goetz et al. 2004). The early diagnosis of PD uses the Hoehn and Yahr scale or the unified Parkinson's disease rating scale (UPDRS). But there are still many issues waiting to be resolved. First, patients rate themselves on the two aforementioned scales based on their own health conditions, leading to a lack of consistency and objectivity.

Gait abnormality is one of the most common symptoms of PD. Current research on the early discovery of gait abnormality in PD is divided into three methods: imaging analysis, intelligent signal processing, and pattern recognition. The threshold of signal method involves the calculation of a single indicator such as step frequency, stride length, walking speed and others to determine whether it falls in a range that is associated with gait anomalies. However, accurately determining a patient's gait condition using a single indicator is difficult. And the threshold for identification of an anomaly can easily vary with the number of samples. The accuracy of detection by this method would be hardly improved (Wu et al. 2017; Ly et al. 2017; Navya et al. 2018).

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Images of the patient are typically captured at various angles using many high-quality cameras. Corresponding nodes on the body are identified to calculate the continuous gait condition of the patient during the test. Although this method can completely record the patient's process of test, the complexity of the image calculation is high, requiring a special room to be set up with dedicated medical staff and testing personnel. The method imposes high medical costs and so is not favorable (Pham 2017; Thuc et al. 2017; Elkholy et al. 2019).

When artificial intelligence is used to analyze gait signals, data processing is involved. In this method, a continuous gait pressure sensing signal is calculated according to abnormal gait conditions (such as gait freezing, stride interval, number of steps and others). The information of abnormal gait conditions of that occurred during the time is recorded as trainable features. Finally, an algorithm is used to find patterns in the obtained data to identify an abnormal gait. Artificial intelligence-based detection has a higher recognition rate and reliability than traditional single-threshold detection (Tahafchi et al. 2017; Baghdadi 2018; Nazmi et al. 2018).

In recent years, most studies of intelligent gait detection for PD have used a single algorithm to train a model. In this work, explainable learning architecture is used in the early diagnosis of PD. In data preprocessing, the dimension reduction method is used in learning architecture to reduce the noise in the dataset. ANOVA with recursive reduction (ARR) is used to improve the dimension reduction. Medical research is strictly regulated, so obtaining relevant data is difficult. Therefore, data imbalance learning is incorporated into the architecture, increasing the effectiveness of processing. In the recognition stage, machine learning is combined with deep learning to increase the accuracy of a training model, greatly increasing its stability. The study mainly focuses on following points.

- This study develops explainable learning architecture to improve the performance of a training model. This explainable architecture clearly shows the processing of each sensing data to help understand system information.
- From the characteristics of sensor data, differences between healthy subjects and PD patients are analyzed. All of the twenty-two features are extracted and divided into three domains—force domain, peak domain and abnormality domain.
- To solve the problem of imbalanced dataset, a synthetic minority oversampling technique (SMOTE) is added to the learning architecture, improving credibility.
- The validity and stability of the model are significantly improved by using ARR to solve the problem of data selection.

2 Literature review

2.1 Gait detection for PD

Gait detection is a very popular theme in the study of PD. Gait detection in the study of PD, as presented in recent research, is performed using three methods, which are based on threshold of collected sensing information, image recognition and artificial intelligence. Rovini et al. (2017) consolidated 136 studies of the use of sensors in the early detection of PD and found a lack of consistency between rating scale and high-cost radiation detection. Owing to small size, low cost, non-invasiveness, ability to record a patient's motor behavior, and wearability, sensors have become important tools for evaluate PD with value in development and research (Aich et al. 2017; Wang et al. 2017; Das et al. 2017).

Most recent research into gait detection in cases of PD has used the threshold method. Most of signals that are collected by sensors (such as accelerometers, gyroscopes and pressure sensors) are used to determine whether gait is normal. Abnormal indicator values suggest abnormalities in the gait of patients with PD (Mileti et al. 2018; Yoneyama et al. 2013). Pham et al. (2017) proposed placing accelerometers on the shank, thigh and lower back and calculating features that are associated with gait behavior (such as mean, standard deviation, variation and freezing of the gait (FoG)), before using statistical methods to find a single feature that falls its normal range. Finally, an intra-class correlation coefficient (ICC) and confusion matrix are used to evaluate similarities among them and to identify features.

A patient's gait cannot easily be accurately determined using a single indicator, and the threshold value of sensing information easily varies with the number of samples, reducing the accuracy of detection. In recent years, many studies have used image recognition for gait detection in PD patients with multiple high-quality cameras, capturing continuous motion pictures of patients from various angles during gait testing. A program is used to convert a picture of a human body into an image that comprises nodes, which are used to calculate the gait condition of patient at the time of testing (Procházka et al. 2014; Sun et al. 2018).

Although image recognition can be used to analyze completely the process of testing, the cost and computational complexity of the construction of system are high. And it requires that a hospital has an examination room with professional medical personnel and testing personnel. Krajushkina et al. (2018) found that the associated costs are high, and the method is therefore difficult to use in a clinic.

2.2 Artificial intelligence analysis

The use of artificial intelligence in gait detection solves the problem of traditional threshold setting and improves the

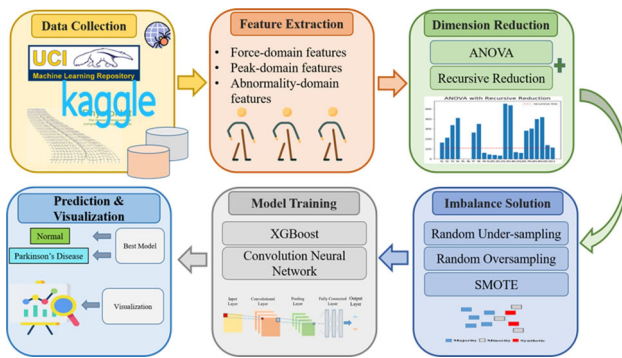


Fig. 1 Proposed explainable learning architecture

results of identification. Analyzing gait signals patients with PD during testing enables features of an abnormal gait to be identified. Learning algorithm is used to identify patterns from features for the purpose of identifying an abnormal gait. Many recent studies have combined AI with gait detection.

Selzler et al. (2018) proposed to measure the gait motion of patients with PD, Huntington's disease and Amyotrophic Lateral Sclerosis using force sensitive resistors. Data files were generated over five minutes and a sampling frequency of 300 Hz. All data were labeled by manual. The features included stance, swing, and stride intervals. The work evaluated performance using K-nearest neighbor (KNN) and random forest algorithms. Gait identification using artificial intelligence has higher precision and reliability than traditional single threshold detection (Yan et al. 2020; Sánchez-Delacruz et al. 2014).

3 Proposed explainable learning architecture

The proposed explainable learning system architecture comprises has parts, corresponding to Data Collection, Feature Extraction, Dimension Reduction, Imbalance Solution, Model Training and Prediction and Visualization, as shown in Fig. 1.

In Data Collection, all data are obtained from PhysioNet using a web crawler. In Feature Extraction, twenty-two features are extracted from the data for normal and PD subjects and divided into three domains: the force domain, the peak domain and the abnormality domain. Dimension Reduction reduces the dimensionality of the dataset, filtering out noise and increasing the detection accuracy of training model. Imbalance Solution adds new data to the original data for normal subjects to reduce the effect of data imbalance in the training model. Model Training uses machine learning and the deep learning algorithm to generate a detection model for PD. The training dataset filters out any effect of noise on testing data, to increase detection accuracy and to reduce

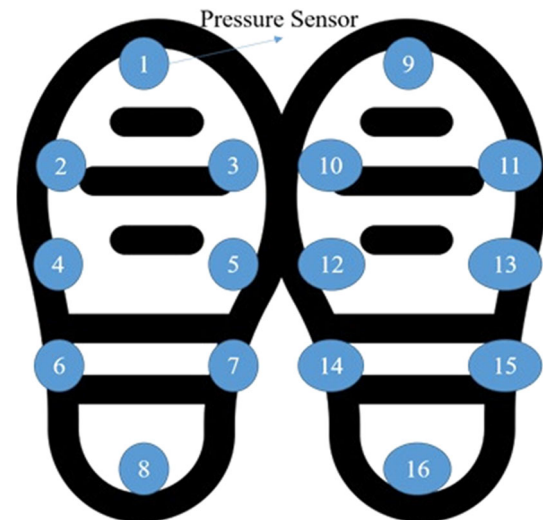


Fig. 2 Location of the pressure sensors

training time. Prediction and Visualization displays testing results and performance.

3.1 Data Collection

In the study, sixteen pressure sensors are placed on the left and right soles of the subjects' shoes, collected data as shown in Fig. 2. The sampling frequency of pressure sensors was 100 Hz. A web crawler is used to collect an information. It sends a GET request to a server and receives a response in the form of an HTML file.

The dataset was obtained for 93 respondents with PD and 73 healthy people. The average age of healthy respondents and those with PD was 66.3. Of the people in the database 59 percent were male. The dataset concerns people who were walking normally or do something while walking. In the do something while walking task, subjects performed seven subtractions while walking; normal walking involved walking for two minutes. Data pertaining to the do something while walking task and outliers were filtered out, leaving training data in 102 files for 40 healthy subjects and 62 PD subjects.

3.2 Feature Extraction

This work extracts twenty-two features based on data characteristics that analyze the differences between healthy people and PD patients. Obtain the average, minimum, maximum and standard deviation of the pressure sensors at sensor positions 1 and 9 every 5 seconds during the walking task. A high-pass filter can filter out the noise in the data, reducing the range of oscillation of signal. Peak detection can locate the local maximum and local minimum of signal data which are important in a signal detection. When the value of signal exceeds an earlier and in the next moment, it may be a

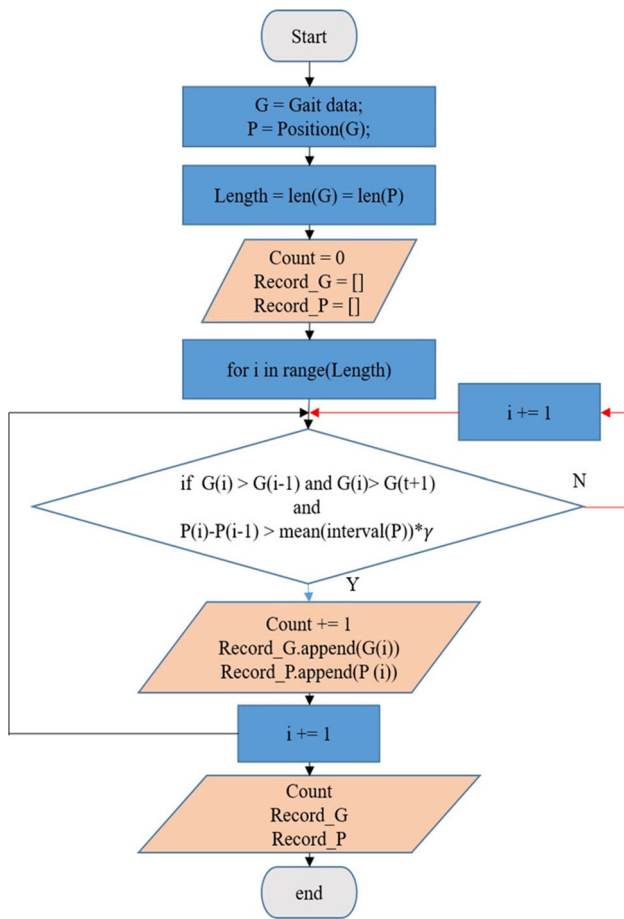


Fig. 3 Process flow of peak detection

peak. In this study, the distance between consecutive peaks is used to determine whether a signal is a peak. Peak detection process flow is shown in Fig. 3.

In the process of peak detection, G is gait data from pressure sensors; P is the position of sensor. Peak domain is extracted using a high-pass filter with a cutoff frequency of 0.05 Hz and peak detection. Besides *Mean*, *Standard Deviation*(SD), *Max* and *Min* value of peak data, the count and interval time (IT) are also expressed using formulas.

$$Count = \sum_{pt=1}^{N_p} x_p \quad (1)$$

$$IT = \sum_{p=1}^{N_p} T_p - T_{p-1} \quad (2)$$

x_p is the pt -th peak data from total number of data; N_p is the total number of peaks; T_p is a pt -th position of time point of x_{pt} ; N_{pt} is the total number of T_p .

PD patients have obvious symptoms that distinguish them from healthy humans. The common gait-related symptoms

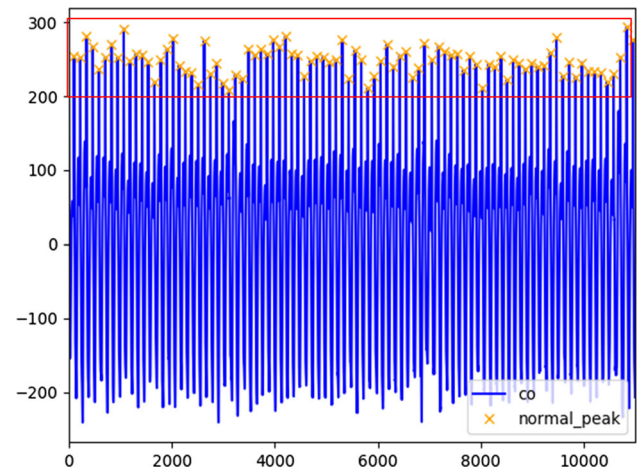


Fig. 4 Pressure data of healthy subject

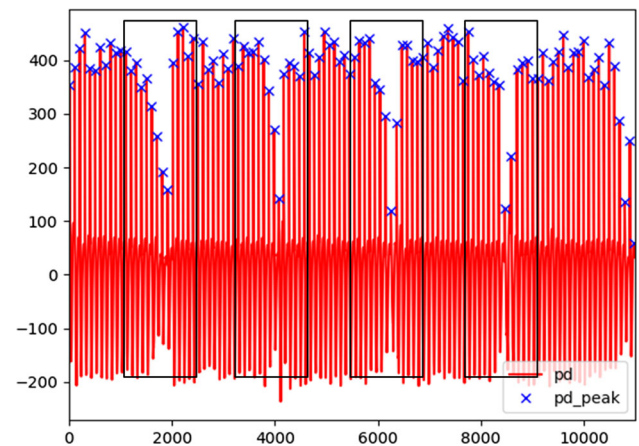


Fig. 5 Pressure data of Parkinson's disease subject

of PD patients include freezing of the gait, short stride length and festinating gait. The difference between healthy and PD subjects can be observed using visualization technology. The gait of healthy subjects is more stable than that of PD subjects, as shown in Figs. 4 and 5.

The abnormality-domain features are used for abnormality detection to identify an abnormal gait. All subjects have different average values of sensor data, to ensure accuracy of detection. When average value was used to the threshold directly, some mistakes were discovered. To correct the mistakes, constants γ_{p1} , γ_{p2} and γ_{p3} are added to equation. They can adjust the threshold of $IT_{abnormal}$ and $PV_{abnormal}$. Abnormality-domain features are expressed using the following formulas.

$$\begin{cases} IT_{abnormal} = \frac{\sum_{p=1}^N (T_p - T_{p-1})}{N} \\ \text{if } (IT_{abnormal} * \gamma_{p2} \leq (T_p - T_{p-1})) \\ ((T_p - T_{p-1}) \leq IT_{abnormal} * \gamma_{p1}), \\ \text{then } N_{abIT} = N_{abIT} + 1 \end{cases} \quad (3)$$

Table 1 Force-domain features

ID	Feature name	Feature description
f1 f2	$Mean_{L\&R}$	The average value of walking task every 5 seconds; the data are from position 1 and 9 of pressure sensors
f3 f4	$Max_{L\&R}$	The maximum of walking task every 5 seconds; the data are from position 1 and 9 of pressure sensors
f5 f6	$Min_{L\&R}$	The minimum of walking task every 5 seconds; the data are from position 1 and 9 of pressure sensors
f7 f8	$Std_{L\&R}$	The standard deviation of walking task every 5 seconds; the data are from position 1 and 9 of pressure sensors.
f9 f10	$N_{PL\&PR}$	The number of peaks every 5 seconds; the data are from position 1 and 9 of pressure sensors; it is calculated through a high-pass filter and peak detection.
f11 f12	$IT_{PL\&PR}$	The interval time every 5 seconds; the data are from position 1 and 9 of pressure sensors; it is calculated through a high-pass filter and peak detection.
f13 f14	$Min_{PL\&PR}$	The minimum of walking task every 5 seconds; the data are calculated through a high-pass filter, and it is from position 1 and 9 of pressure sensors.
f15 f16	$Std_{PL\&PR}$	The standard deviation of walking task every 5 seconds; the data are calculated through a high-pass filter, and it is from position 1 and 9 of pressure sensors.
f17 f18	$Max_{PL\&PR}$	The standard deviation of walking task every 5 seconds; the data are calculated through a high-pass filter, and it is from position 1 and 9 of pressure sensors.
f19 f20	$Mean_{PL\&PR}$	The mean of walking task every 5 seconds; the data are calculated through a high-pass filter, and it is from position 1 and 9 of pressure sensors.
f21 f22	$N_{abL\&abR}$	The number of occurrences of abnormal status every 5 seconds; the data are from position 1 and 9 of pressure sensors, and it is calculated through a high-pass filter and peak detection.

$$\begin{cases} PV_{abnormal} = \frac{\sum_{pv=1}^{N_{pv}} (x_{pv} - x_{pv-1})}{N_{pv}}, \\ \text{if } |(x_{pv} - x_{pv-1})| \geq PV_{abnormal} * \gamma_{p3} \\ \text{then } N_{abPV} = N_{abPV} + 1 \end{cases} \quad (4)$$

$$N_{ab} = N_{abIT} + N_{abPV} \quad (5)$$

γ_{p1} and γ_{p2} are constants that can adjust the threshold ($T_p - T_{p-1}$); x_{pv} is the pv-th data of peaks gait; N_{pv} is the total number of x_{pv} ; γ_{p3} is a constant that can adjust the threshold of $|(x_{pv} - x_{pv-1})|$. In this study, γ_{p1} , γ_{p2} and γ_{p3} are 0.8, 1.7 and 2.4, respectively. The detailed description of features is shown in Table 1

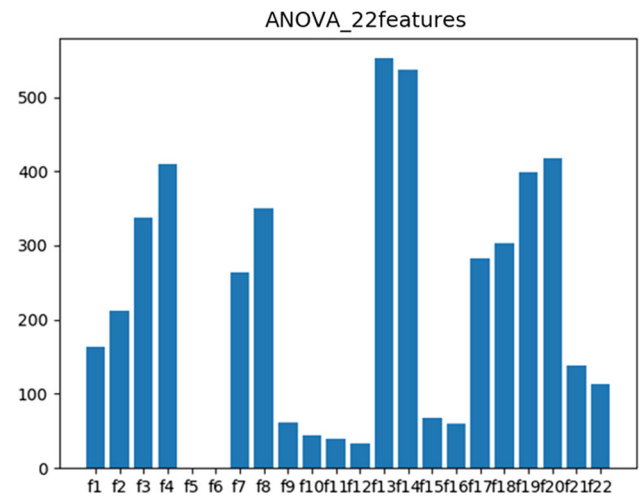


Fig. 6 F-value of ANOVA for 22 features

3.3 Dimension reduction

An excessive number of features will increase the training time and worsen the performance of training model. Dimension reduction reduces the number of dimensions of the dataset, filtering out noise in the dataset and improve the detection accuracy of the model. All twenty-two features were evaluated using Analysis of Variance (ANOVA). ANOVA method may be in comparison between more than two samples. It compares these categories for a variable measured on an equidistant/ratio scale, whether the difference in the mean is significant (Rutherford 2001; Rouder et al. 2016). The F-value represents the importance of a feature in ANOVA. All features have an F-value, as shown in Fig. 6. The values for features f5 and f6 approach zero so these can be removed directly from all features; however, features f9, f10, f11, f12, f15 and f16 have lower values than other features and whether these should be removed cannot be determined.

Therefore, to solve problem of feature selection, the use of ARR is proposed (Hesthaven and Zhang 2016; Xu et al. 2002). The process flow of ARR is shown in Fig. 7. In the process of ARR analysis, an F-value represents the importance of a feature. First, a feature is deleted from the dataset when its F-value is less than one. Second, a set of features is deleted sequentially from a dataset using a recursive method when their F-values are less than threshold of ARR. Third, the remaining features are used to constitute a new dataset, all of which will be used to train a model. ARR threshold can be expressed as shown in the following formula.

$$ARR = \frac{\sum_{v=1}^{N_f} F_v}{N_f} * \gamma_{p4} \quad (6)$$

F_v is the v-th F-value; N_f is the total number of features; γ_{p4} is a constant that can control the threshold of ARR. When

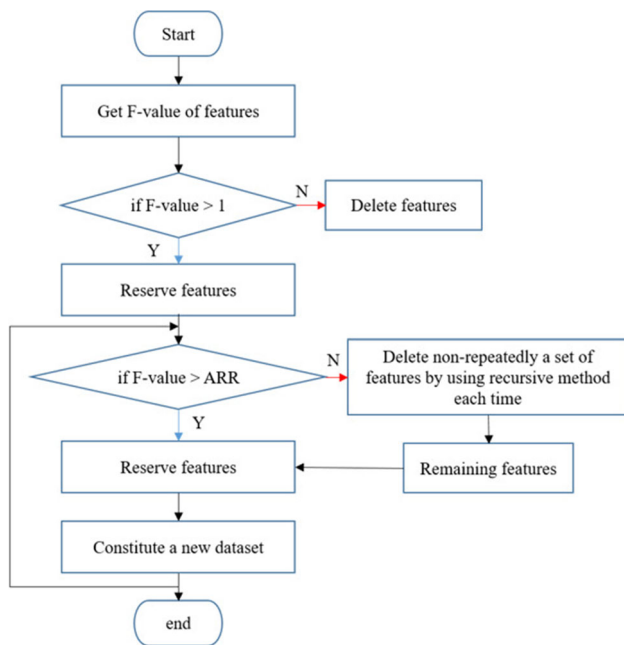


Fig. 7 The flowchart of ARR analysis

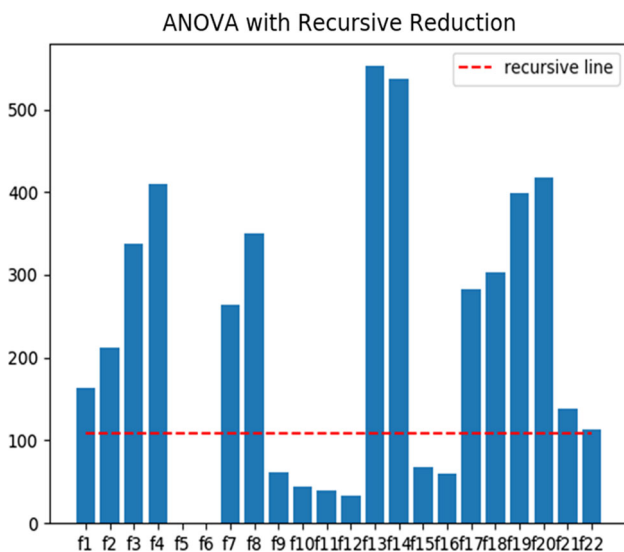


Fig. 8 ARR analysis result

the F-value of features is less than ARR, this feature will not be used. ARR analysis is shown in Fig. 8.

3.4 Imbalance Solution and Model Training

In the process of training a model, an imbalanced dataset allows a majority category to have a larger weight than a minority category, directly affecting the performance of the training of the model. In this work, gait data for PD and healthy subjects exhibit the imbalance problem. In the dataset, the number of PD subjects exceeds the number

of healthy subjects, affecting the accuracy of the detection model.

In recent years, most studies of gait detection have appear undersampling, synthetic sample and oversampling to solve the imbalance problem. In this study, Imbalance Solution adds new data for normal subjects to the original data using SMOTE to reduce the effect of data imbalance in the training model. SMOTE is one of the methods commonly used to solve the problem of category imbalance (Fernández et al. 2018; Han et al. 2005; Douzas and Bacao 2019). The method is to randomly select a sample of a few categories as a seed sample. Then, it finds out that the surrounding samples are of the same minority category and choose one as the selected sample. And it generates a dummy sample between the two minority class samples.

Model Training uses machine learning and the deep learning algorithm to develop a detection model for cases of PD. The training dataset is processing from Data Collection, Feature Extraction, Dimension Reduction and Imbalance Solution. The above processing filters out the influence of noise on the collection dataset, improves detection accuracy and reduces training time. The machine learning and deep learning are used to train a model for gait detection for cases of PD. In the process of training, batch normalization combined with convolutional neural network (CNN) architecture (Acharya et al. 2017; Li et al. 2016) is to improve the stability and performance. The overall flowchart is shown in Fig. 9, including the previous processes of Data Collection, Feature Extraction, Dimension Reduction, Imbalance Solution, Model Training, and Prediction and Visualization.

4 Experiments and analysis

This section presents an analysis of the performance of explainable learning architecture, which is proposed visualizes relevant results. The machine learning algorithms are written by Python 3.6. The deep learning algorithm is written by the architecture of TensorFlow-gpu 1.8, which is based on Python 3.6. The details of specifications of the hardware and software configuration are shown in Table 2, respectively. In the early stage of Dimension Reduction, ANOVA was used to evaluate the importance of features; however, whether some features should be removed from all features could not be determined even though their F-values were far lower than others; these features were f9, f10, f11, f12, f15 and f16.

To solve imbalanced dataset problems, imbalance learning is combined with ARR mechanism in an explainable learning architecture. A model for gait detection for patients with PD was trained using the XGBoost tool. XGBoost is an optimized distributed gradient boosting library designed to be highly efficient, flexible and portable. It implements machine learning algorithms under the gradient boosting framework.

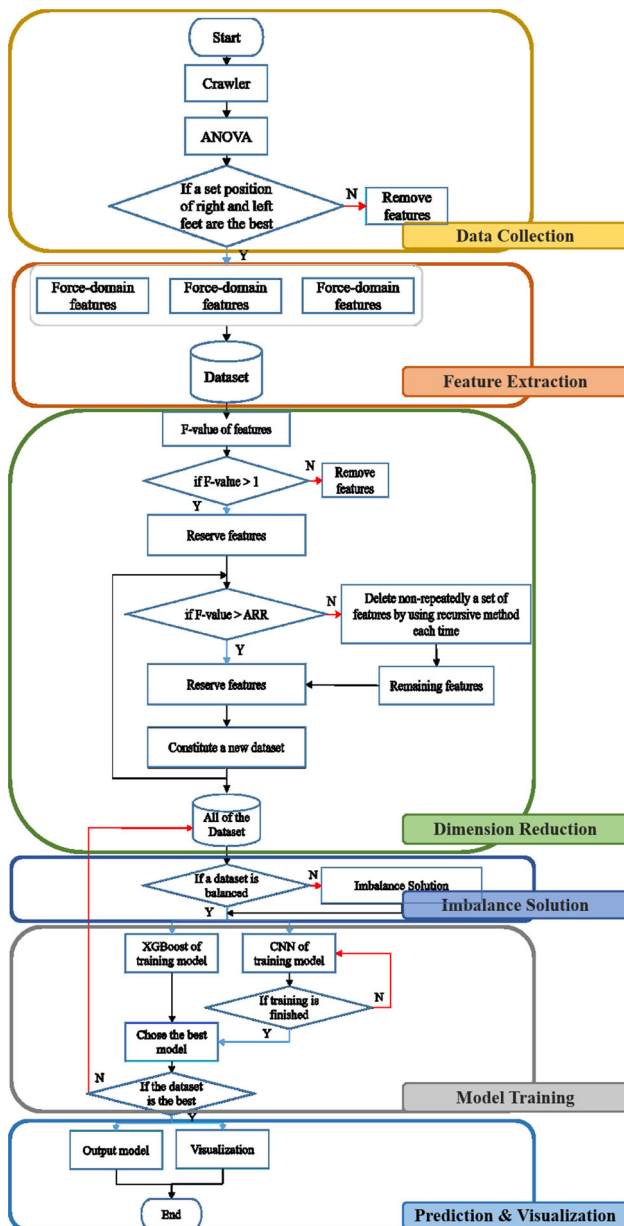


Fig. 9 Flowchart of proposed system

No correlation was found among features, so max-pooling was not used during training, and the size of the filter was $1 \times 1 \times 1$, representing a diversity of features.

Accuracy, precision, recall, standard deviation and F1-score are used to evaluate the performance of the training model. The definition of accuracy, precision, recall and F1-score is shown as below.

$$Precision = \frac{TP}{TP + FP} \quad (7)$$

$$Recall = \frac{TP}{TP + FN} \quad (8)$$

Table 2 Specifications of the hardware and software

Type	Specification
Central processing unit	Intel(R) Core(TM) i5-7600K CPU @ 3.8 GHz
Memory	16 GB
Operating system	Windows 10
Graphics processing unit	GIGABYTE GeForce GTX 1080 Ti * 1
Hard disk drive	1 TB
Solid-state disk	256 GB
CUDA	v9.0
cuDNN	v7.1
Python	3.6
TensorFlow-gpu	v1.8

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (9)$$

$$F1 - score = \frac{2PR}{P + R} = \frac{2 * TP}{2 * TP + FP + FN} \quad (10)$$

The confusion matrix, line graph of accuracy and line graph of loss are used to visualize the results. A confusion matrix represents the prediction performance. All of the collected data are divided into two categories: the label 0 represents a healthy person and represents a PD patients.

4.1 Comprehensive comparison of explainable learning

This work uses confusion matrix to visualization of accuracy, precision, recall, F1-score and SD of performance of training model for gait detection in cases of PD. A summary of explainable learning architecture is shown in Fig. 9, including a comparison of the use and non-use of Imbalance Solution. The training time with XGBoost is much shorter than CNN training, because XGBoost avoids convolution calculation of related data. In accuracy, precision, recall and F1-score, the performance of XGBoost with its computing module is generally better than that of CNN computing. However, CNN can get the best performance when collocated with ARR and SMOTE. This result shows that the ARR calculation result has a greater impact on the CNN algorithm.

4.2 Comparison of different studies

Assessment indexes are used to compare the training model herein with those in other studies. The performance analysis demonstrates that the learning architecture in this study outperforms those in other studies. Mittra and Rustagi (2018) used a dataset from PhysioNet to train a model for gait

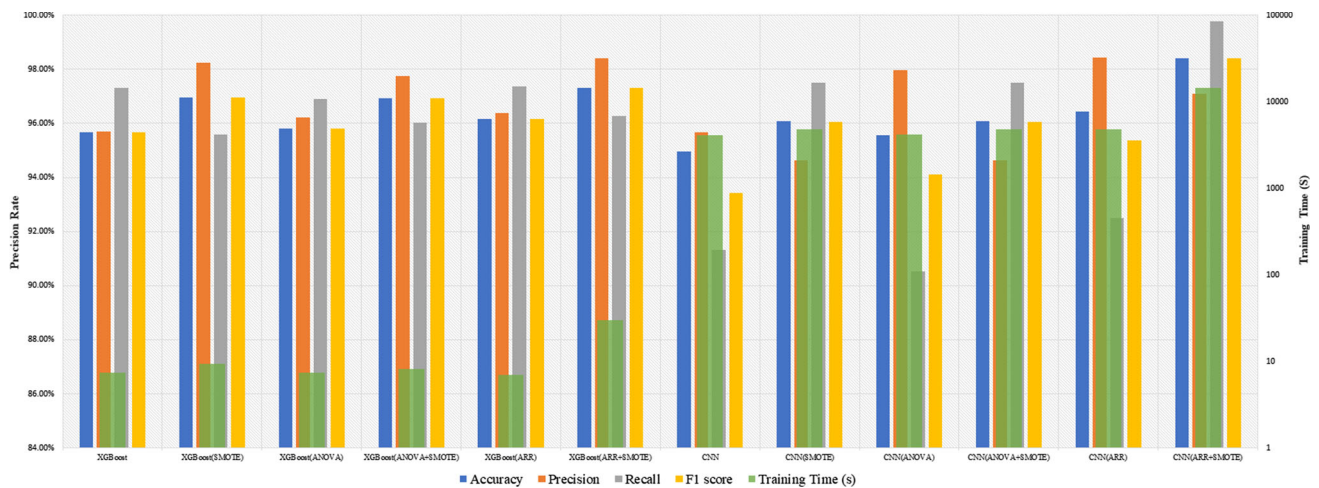


Fig. 10 Comprehensive comparison of explainable learning

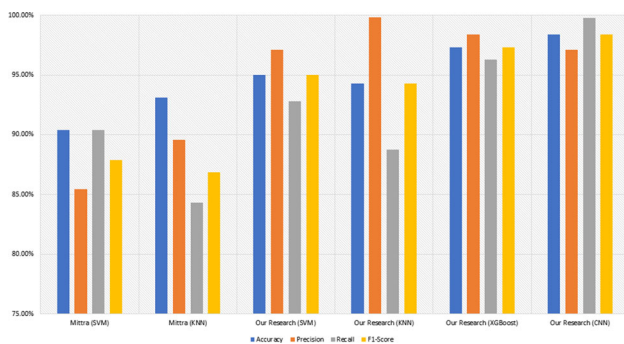


Fig. 11 Comparison results of different studies

detection in cases of PD. They used a different method of feature extraction from that used herein, but used support vector machine (SVM) and KNN algorithms to train their model.

The results of comparison are shown in Fig 10. The indexes of performance for the explainable learning architecture in this work are higher than in other papers. This work establishes the effectiveness and stability of the proposed learning architecture (Fig. 11).

5 Conclusions

With the combined application of learning computing and sensing technology, it can be more effectively used in smart healthcare. The explainable learning architecture is proposed to solve the problem of the early diagnosis of PD by analyzing differences between subjects with PD and healthy subjects. The research adapted three domains, which are the force domain, the peak domain and the abnormality domain to extract twenty-two features. The ARR is proposed to made regarding whether features should be removed. Finally,

according to the test results, this research reduces the number of sensors from 16 to 2 and finds 6 key features.

The learning model increases the accuracy of the performance analysis from 95.68% to 97.32% by integrating with XGBoost and increases the accuracy of the performance analysis from 94.96% to 98.4% by integrate with CNN. The validity and stability of the proposed model are thus improved significantly. In the future, the study will gradually explore the differences corresponding to gait detection in different stages of PD.

Declarations

Conflict of interest We declare that we do not have any commercial or associative interest, that represents a conflict of interest in connection with the work submitted.

Appendix

See Tables 3 and 4.

Table 3 Table of abbreviations

Abbreviations	Full name
ANOVA	Analysis of variance
ARR	ANOVA with recursive reduction
FoG	Freezing of the gait
ICC	Intra-class correlation coefficient
KNN	K-nearest neighbor
PD	Parkinson's disease
SMOTE	Synthetic minority oversampling technique
SVM	Support vector machine

Table 4 Table of symbols definitions

Symbols	Definitions
x_o	Pressure sensing signal
N_0	Total number of data
μ_o	Average value of sensor data
x_p	p-th Sensing pressure data from sensor
N_p	Total number of data
μ_p	Average of x_p
X_{pt}	pt-th Peak data form N_p
N_{pt}	Total number of peaks
x_{ptime}	ptime-th Position of time point of X_{pt}
N_{ptime}	Total number of x_{ptime}
γ_{p1}	Constant that can adjustment the threshold
γ_{p2}	Constant that can adjustment the threshold
x_{pv}	pv-th Data of peaks gait
N_{pv}	Total number of x_{pv}
γ_{p3}	Constant that can adjustment the threshold
x_{fv}	fv-th F-value
N_{pv}	Total number of features
γ_{p4}	Constant that can control the threshold of ARR

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